

Ring switching, generalized

How to transform a PCS for $\mathbb{F}_{2^f}$ into a PCS for $\mathbb{F}_{2^e}$, for arbitrary e and f .

Goal. We want f *bit*-valued multilinear $P_1, \dots, P_f : \{0, 1\}^m \rightarrow \{0, 1\}$ over the same m variables, to be committed through a dense PCS over the *packing* field $\mathbb{F}_{2^f}$ (if we can commit to bits, we can trivially commit in any extension). The evaluation claims, however, arrive over a *different* field: at an opening point $r \in \mathbb{F}_{2^e}^m$ we are given $P_i(r) = \alpha_i \in \mathbb{F}_{2^e}$. Ring switching [1] packs the f polynomials into one $\mathbb{F}_{2^f}$ -dense polynomial and reduces the f claims to a single claim on it. The two extension degrees, f (packing) and e (opening), are unrelated (not necessarily a sub-field / extension-field).

Packing. Fix an $\mathbb{F}_2$ -basis $\{b_i^F\}_{i=1}^f$ of $\mathbb{F}_{2^f}$. Consider the multilinear polynomial:

$$P_{\text{packed}}(y) = \sum_{i=1}^f P_i(y) b_i^F, \quad y \in \{0, 1\}^m,$$

dense over $\mathbb{F}_{2^f}$, which we commit to.

Reduction. The claims share a *common* point $r \in \mathbb{F}_{2^e}^m$; each is

$$\alpha_i = P_i(r) = \sum_{y \in \{0, 1\}^m} \text{eq}(r, y) P_i(y),$$

with bits $P_i(y)$ and weights $\text{eq}(r, y) \in \mathbb{F}_{2^e}$. Fix an $\mathbb{F}_2$ -basis $\{b_w^E\}_{w=1}^e$ of the *opening* field $\mathbb{F}_{2^e}$ and expand the weights, $\text{eq}(r, y) = \sum_{w=1}^e A(y, w) b_w^E$, with $A : \{0, 1\}^m \times [e] \rightarrow \mathbb{F}_2$ public, multilinear in y for each fixed w , with efficiently computable MLE (see [Evaluating \$B\$](#) below). Substituting and pulling out the basis,

$$\alpha_i = \sum_{w=1}^e b_w^E \underbrace{\sum_{y \in \{0, 1\}^m} A(y, w) P_i(y)}_{\in \mathbb{F}_2},$$

so the w -th $\mathbb{F}_2$ -coordinate of α_i is $\alpha_{w,i} = \sum_{y \in \{0, 1\}^m} A(y, w) P_i(y)$. Now recombine *across* i with the *packing* basis $\{b_i^F\}$: define, for each $w \in \{1, \dots, e\}$,

$$t_w := \sum_{i=1}^f \alpha_{w,i} b_i^F = \sum_{y \in \{0, 1\}^m} A(y, w) \underbrace{\sum_{i=1}^f P_i(y) b_i^F}_{= P_{\text{packed}}(y)} = \sum_{y \in \{0, 1\}^m} A(y, w) P_{\text{packed}}(y) \in \mathbb{F}_{2^f}$$

one claim per w , now purely about the packed polynomial P_{packed} over $\mathbb{F}_{2^f}$ and public A . Batch these e claims with powers of a random $\gamma \leftarrow \mathbb{F}_{2^f}$ into a single sumcheck on $\sum_{y \in \{0, 1\}^m} B(y) P_{\text{packed}}(y) = \sum_{w=1}^e \gamma^w t_w$, with $B(y) = \sum_{w=1}^e \gamma^w A(y, w)$. Its output is the packed claim $P_{\text{packed}}(r') = \alpha'$ at a point $r' \in \mathbb{F}_{2^f}^m$ (plus a $B(r')$ the verifier computes itself). Done.

Soundness. Suppose the input is wrong: $\alpha_i \neq P_i(r)$ for some i , against the committed P_{packed} . Then the reduction outputs a correct packed claim $P_{\text{packed}}(r') = \alpha'$ with probability at most

$$\underbrace{\frac{e}{2^f}}_{\text{batching}} + \underbrace{\frac{2m}{2^f}}_{\text{sumcheck}} = \frac{2m + e}{2^f},$$

Proof:

1. The map $(\alpha_i)_{i=1}^f \mapsto (t_w)_{w=1}^e$ is a *bijection* $\mathbb{F}_{2^e}^f \rightarrow \mathbb{F}_{2^f}^e$. So a wrong input forces, with certainty, $t_w \neq \sum_{y \in \{0,1\}^m} A(y, w) P_{\text{packed}}(y)$ for some w .
2. By Schwartz–Zippel, $\sum_{w=1}^e \gamma^w t_w$ then differs from $\sum_{y \in \{0,1\}^m} B(y) P_{\text{packed}}(y)$ except with probability $\leq e/2^f$ over $\gamma \leftarrow \mathbb{F}_{2^f}$.
3. Sumcheck started from this false value (summand of degree 2, over m variables) ends at a correct claim $P_{\text{packed}}(r') = \alpha'$ with probability $\leq 2m/2^f$.

Evaluating B

It remains to compute $B(r')$, where $B(y) = \sum_{w=1}^e \gamma^w A(y, w)$ and $A(y, w) \in \mathbb{F}_2$ is the w -th coordinate of $\text{eq}(r, y)$, i.e. $\text{eq}(r, y) = \sum_{w=1}^e A(y, w) b_w^E$. Following [2], this gives B a closed form on the hypercube. Let $\Phi : \mathbb{F}_{2^e} \rightarrow \mathbb{F}_{2^f}$ be the $\mathbb{F}_2$ -linear map with $\Phi(b_w^E) = \gamma^w$. For every $y \in \{0, 1\}^m$:

$$\Phi(\text{eq}(r, y)) = \Phi\left(\sum_{w=1}^e A(y, w) b_w^E\right) = \sum_{w=1}^e A(y, w) \Phi(b_w^E) = \sum_{w=1}^e A(y, w) \gamma^w = B(y).$$

So on the hypercube B is “compute $\text{eq}(r, y) \in \mathbb{F}_{2^e}$, then apply Φ ”.

Closed form. Two elementary facts do all the work.

Fact 1 (eq in characteristic 2). In each factor of eq the two $x_j y_j$ terms cancel:

$$\text{eq}(x, y) = \prod_{j=1}^m (x_j y_j + (1 + x_j)(1 + y_j)) = \prod_{j=1}^m (1 + x_j + y_j).$$

Fact 2 (linear maps are sums of squarings). View Φ as mapping into $L = \mathbb{F}_{2^{\text{lcm}(e, f)}}$, the smallest field containing both $\mathbb{F}_{2^e}$ and $\mathbb{F}_{2^f}$ (for $e = f$: $L = \mathbb{F}_{2^e}$, nothing to embed). Squaring is additive in characteristic 2, so any *linearized polynomial* [3]

$$a \mapsto c_0 a + c_1 a^2 + c_2 a^4 + \cdots + c_{e-1} a^{2^{e-1}}, \quad c_k \in L,$$

is an $\mathbb{F}_2$ -linear map $\mathbb{F}_{2^e} \rightarrow L$. We claim Φ is of this form, and find the c_k by writing out what Φ must do on the basis, $\Phi(b_w^E) = \gamma^w$:

$$\sum_{k=0}^{e-1} (b_w^E)^{2^k} c_k = \gamma^w, \quad w = 1, \dots, e :$$

e linear equations in the e unknowns c_k . The coefficient matrix $((b_w^E)^{2^k})_{w,k}$ is a *Moore matrix*, invertible exactly because the b_w^E are $\mathbb{F}_2$ -linearly independent [4], so the c_k exist and are unique;

the resulting linearized polynomial agrees with Φ on an $\mathbb{F}_2$ -basis, hence (both being $\mathbb{F}_2$ -linear) everywhere. The matrix depends only on the basis, not on γ : invert it once and for all; each reduction then reads off its $(c_k)_{k < e}$ by one matrix-vector product, $O(e^2)$ multiplications.

Now compute. Squaring respects sums and products and fixes bits ($y_j^2 = y_j$), so on the hypercube, by Fact 1,

$$\text{eq}(r, y)^{2^k} = \prod_{j=1}^m (1 + r_j + y_j)^{2^k} = \prod_{j=1}^m (1 + r_j^{2^k} + y_j) = \text{eq}(r^{2^k}, y), \quad r^{2^k} := (r_1^{2^k}, \dots, r_m^{2^k}).$$

Since B is multilinear, evaluating it at r' and expanding with Fact 2:

$$B(r') = \sum_{y \in \{0,1\}^m} B(y) \text{eq}(r', y) = \sum_{y \in \{0,1\}^m} \Phi(\text{eq}(r, y)) \text{eq}(r', y) = \sum_{k < e} c_k \sum_{y \in \{0,1\}^m} \text{eq}(r^{2^k}, y) \text{eq}(r', y).$$

The inner sum evaluates, at r' , the multilinear extension of $y \mapsto \text{eq}(r^{2^k}, y)$, which is $\text{eq}(r^{2^k}, \cdot)$ itself. So it collapses to $\text{eq}(r^{2^k}, r')$, and by Fact 1 again:

$$B(r') = \sum_{k < e} c_k \prod_{j=1}^m (1 + r_j^{2^k} + r'_j).$$

Cost. Level k needs one squaring per coordinate (to get $r_j^{2^k}$ from $r_j^{2^{k-1}}$) and one m -factor product: $O(em)$ multiplications in L , on top of the $O(e^2)$ for the c_k : plain field arithmetic throughout, no bit decompositions.

Remarks

1. *Small $\mathbb{F}_{2^f}$.* If 2^f is too small for the error $(2m + e)/2^f$, sample γ (and run the sumcheck) in an extension $L \supseteq \mathbb{F}_{2^f}$: the error becomes $(2m + e)/|L|$, and P_{packed} is just opened at $r' \in L^m$.
2. *Any field.* Nothing uses characteristic 2: replace $\mathbb{F}_2$ by any field K and $\mathbb{F}_{2^f}, \mathbb{F}_{2^e}$ by degree- f and degree- e extensions of K ; the reduction still works, and efficient algorithms for evaluating B exist in general too.

Credits

The idea comes from Lev Soukhanov, *[[alloc]init]*.

References

- [1] Benjamin E. Diamond and Jim Posen. Polylogarithmic proofs for multinears over binary towers. In *Advances in Cryptology – EUROCRYPT 2026*, pages 3–32. Springer, 2026.
- [2] Benedikt Bünz, Ron Rothblum, and William Wang. Flock: Fast proving for batch Boolean computations. Cryptology ePrint Archive, Paper 2026/1329, 2026. <https://eprint.iacr.org/2026/1329>.
- [3] Wikipedia. Linearised polynomial. https://en.wikipedia.org/wiki/Linearised_polynomial.

[4] Wikipedia. Moore matrix. https://en.wikipedia.org/wiki/Moore_matrix.